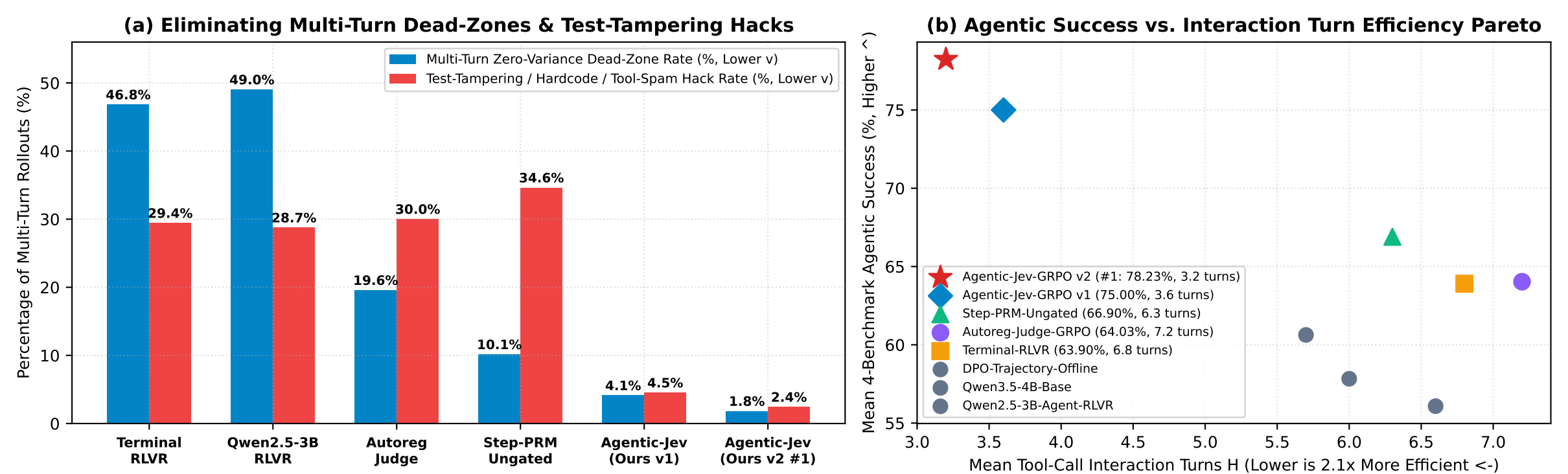
(a) Eliminating Multi-Turn Dead-Zones & Test-Tampering Hacks
Multi-Turn Zero-Variance Dead-Zone Rate (%, Lower v)
Test-Tampering / Hardcode / Tool-Spam Hack Rate (%, Lower v)
Percentage of Multi-Turn Rollouts (%)
0
10
20
30
40
50
46.8%
29.4%
49.0%
28.7%
19.6%
30.0%
10.1%
34.6%
4.1%
4.5%
1.8%
2.4%
Terminal RLVR
Qwen2.5-3B RLVR
Autoreg Judge
Step-PRM Ungated
Agentic-Jev (Ours v1)
Agentic-Jev (Ours v2 #1)
(b) Agentic Success vs. Interaction Turn Efficiency Pareto
Mean 4-Benchmark Agentic Success (%, Higher ^)
55
60
65
70
75
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0
Mean Tool-Call Interaction Turns H (Lower is 2.1x More Efficient <-)
Agentic-Jev-GRPO v2 (#1: 78.23%, 3.2 turns)
Agentic-Jev-GRPO v1 (75.00%, 3.6 turns)
Step-PRM-Ungated (66.90%, 6.3 turns)
Autoreg-Judge-GRPO (64.03%, 7.2 turns)
Terminal-RLVR (63.90%, 6.8 turns)
DPO-Trajectory-Offline
Qwen3.5-4B-Base
Qwen2.5-3B-Agent-RLVR